AI TrustLens Explainable AI Output Audit Platform Using Evidence-Based Verification

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Abstract:-

Artificial Intelligence (AI) systems such as ChatGPT, Gemini, Claude, and other Large Language Models (LLMs) have become widely used for generating information in domains such as education, software development, research, and technology. Although these systems provide quick and intelligent responses, they may also generate hallucinated, outdated, or unsupported information. Since AI-generated content is often presented with high confidence, users may find it difficult to determine whether the information is reliable. Existing AI systems provide limited transparency regarding the correctness of their responses, creating a need for an independent platform that can verify AI-generated outputs using trusted evidence.

The proposed work is divided into two phases. Phase I focuses on developing an AI output auditing framework capable of extracting factual claims from AI-generated responses and retrieving relevant evidence from trusted knowledge sources. The retrieved evidence will be compared with the generated claims using Natural Language Processing (NLP) and semantic similarity techniques to determine whether each claim is Supported, Contradicted, or Not Enough Evidence. This phase aims to establish an accurate and reliable verification process while evaluating the effectiveness of evidence retrieval and claim verification across selected domains such as Programming, Artificial Intelligence, Computer Science, and Technology.

Phase II focuses on designing and implementing an Explainable AI (XAI) framework that presents verification results in a transparent and user-friendly manner. The platform will generate an explainable audit report containing supporting evidence, confidence scores, claim-wise verification results, and an overall Trust Score to help users understand the reliability of AI-generated content. The proposed framework will be evaluated using suitable performance metrics such as verification accuracy, semantic similarity, response time, and user interpretability. The expected outcome of this work is a practical AI auditing platform that improves the transparency, reliability, and trustworthiness of AI-generated information while promoting responsible AI usage across various real-world applications.